BLUESTAQ™ | NDSU NORTH DAKOTA STATE UNIVERSITY

STRAIGHT ROWS HACKATHON · NOVEMBER 13–14, 2026

Food security is national security.

One drifting tractor is one farm’s bad morning.
The dependency behind it reaches the whole nation.



Earth: NASA / Apollo 17 crew · December 7, 1972

CLOSE TO HOME

North Dakota feeds the nation.

#1 in the U.S. in 2024 for:

- **Spring wheat**
- **Durum wheat**
- **Canola**
- **Sunflowers**
- **Dry edible beans**
- **Flaxseed**

The challenge fleet works near Casselton, in Cass County.

USDA NASS, North Dakota Agricultural Statistics 2025
Image: NASA JSC · ISS Expedition 37 · Sept. 15, 2013



WHY THIS MATTERS

The modern farm runs on space.

58%

of U.S. corn acres planted with auto-steer guidance

USDA ERS · 2016 survey
Up from 5% in 2001

$1.4T

in U.S. private-sector benefits from GPS

RTI for NIST · 1984–2017
2017 dollars

$1B

estimated loss per day in a GPS outage

Up to 50% higher during planting season

When guidance fails, the planting window does not wait.

USDA Economic Research Service · RTI International for NIST, 2019

Meet Bluestaq.

Defense-grade data infrastructure, proven across national security missions, now for every organization.

2018

Founded in Colorado Springs

First built for the Space Force's Space Domain Awareness mission

300+

data types in the Unified Data Library

Bluestaq is its primary developer

200M+

records pulled from the UDL every day

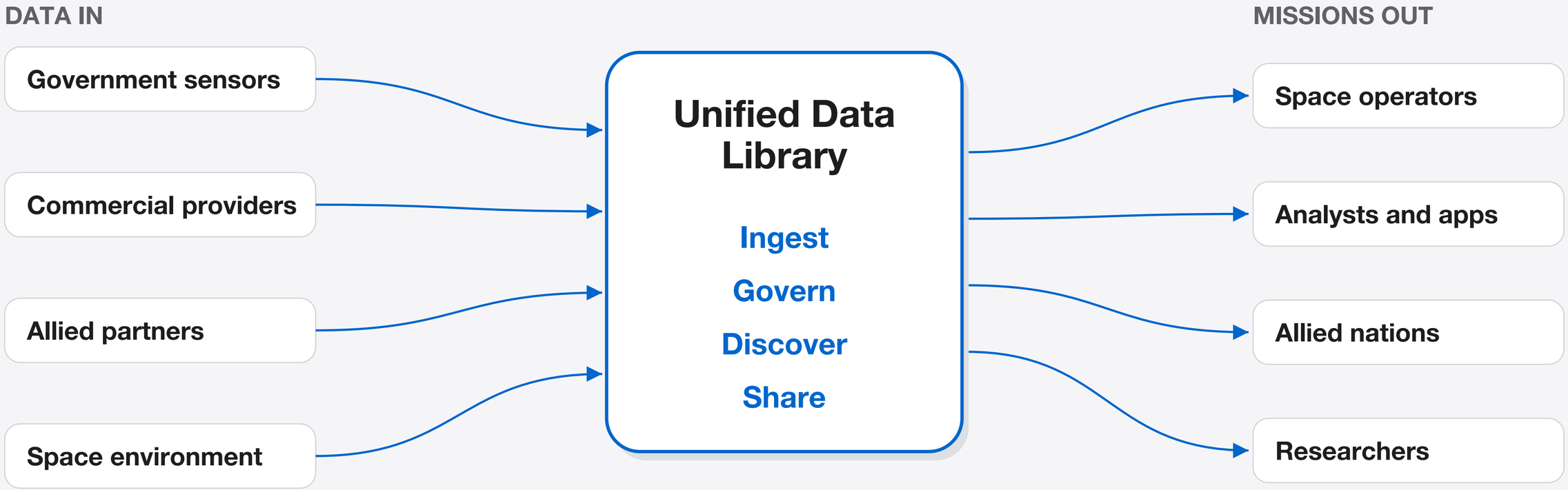
Up to 70 applications 2024 report to Congress

"The question is the same on a farm as on a mission."

Bluestaq About page · Space Systems Command, Mar. 2024 · Breaking Defense, June 2024 · Quote: Bluestaq /ARQ for agriculture, 2026

THE UNIFIED DATA LIBRARY

One library. Many sources. Many missions.



Space Systems Command, Mar. 11, 2024 · Breaking Defense, June 27, 2024 · Bluestaq · Simplified; not every source or user is shown

FROM SIMULATION TO MISSION

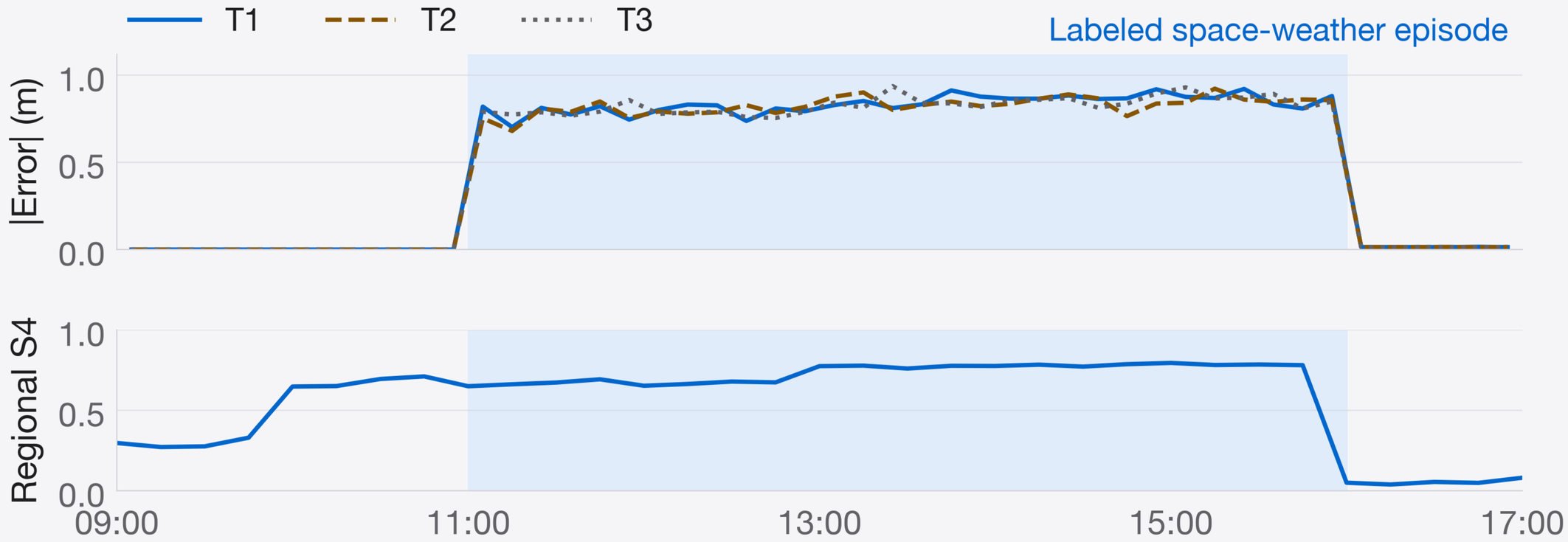
Your challenge, in real-world data.

IN YOUR CHALLENGE		IN THE REAL WORLD
sgi_indices.json Kp and Dst geomagnetic indices	→	**Sgi** Space-weather indices, including Kp and Dst
scintillation_series.json Regional S4 scintillation proxy	→	**IonoObservations** Ionospheric observations · S4 is simplified here
rfi_reports.json Delayed, uncertain interference reports	→	**Emireport** Spectrum-interference reports, including GPS
constellation_advisories.json Notices about satellite status	→	**Notification** Community notices and alerts
tractor_telemetry.csv.gz Satellites, C/N0, fix state, position	→	**GnssObservationset** GNSS receiver observations and status

Different owners, clocks and reliability. Joining them is the job.

Data-type names from Bluestaq’s public UDL SDK (github.com/Bluestaq, Apache-2.0) · All challenge data is simulated

Put the evidence on the same clock.



Public development data · May 5, 2026 (UTC) · Ten-minute medians can hide brief events

THE REPOSITORY

Everything you need, in one place.

bluestaq.github.io/ndsu-hackathon-submissions

Agenda · Get started · Submit · Results · Official guide · Repository: github.com/Bluestaq/ndsu-hackathon-submissions

1 Get the data

AT KICKOFF

Clone the repository, then run

python starter/scripts/get_data.py

The data lands in starter/data/.

2 Build privately

FIRST HOUR

Copy starter/ to your own private repo. Run the baseline, validate, then make one measurable change.

3 Submit one PR

BY SAT 4:00 PM CT

Make your repo public, fork ours, add exactly one file:

submissions/<github-username>/submission.json

GitHub username only. Never your email or student ID. Don’t open your pull request early.

Questions: universityprograms@bluestaq.com

Build your future.

ONE WINNER · BOTH AWARDS

Paid summer internship

Bluestaq HQ
Colorado Springs

Full one-year scholarship

Tuition paid for one year.
Continue your education.

Everything you need: bluestaq.github.io/ndsu-hackathon-submissions

Internship subject to standard eligibility and hiring requirements.